

SEACAR Coral Analysis: Species Richness

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Contents

Important Notes	1
Coral Species Richness	2
Biscayne Bay-Cape Florida to Monroe County Line Aquatic Preserve	2
Coupon Bight Aquatic Preserve	3
Florida Keys National Marine Sanctuary	5
Kristin Jacobs Coral Aquatic Preserve	6
Libraries and Settings	8
File Import	8
Data Filtering	8
Managed Area Statistics	10
Appendix I: Managed Area Species Richness	13

Important Notes

The purpose of this script is to determine species richness by species of grazers and reef-dependent species, create managed area statistics, generate plots, and create reports in pdf form for Coral data.

These scripts were created by [J.E. Panzik \(jepanzik@usf.edu\)](mailto:jepanzik@usf.edu) for SEACAR. Updated by [T.G. Hill \(Tyler.Hill@FloridaDEP.gov\)](mailto:Tyler.Hill@FloridaDEP.gov).

All scripts and outputs can be found on the SEACAR GitHub repository:

https://github.com/FloridaSEACAR/SEACAR_Trend_Analyses

This markdown file is designed to be compiled by [Coral_Plot_Render.R \(https://github.com/FloridaSEACAR/SEACAR_Trend_Analyses/blob/main/Coral/Coral_Plot_Render.R\)](https://github.com/FloridaSEACAR/SEACAR_Trend_Analyses/blob/main/Coral/Coral_Plot_Render.R).

Coral Species Richness

Biscayne Bay-Cape Florida to Monroe County Line Aquatic Preserve

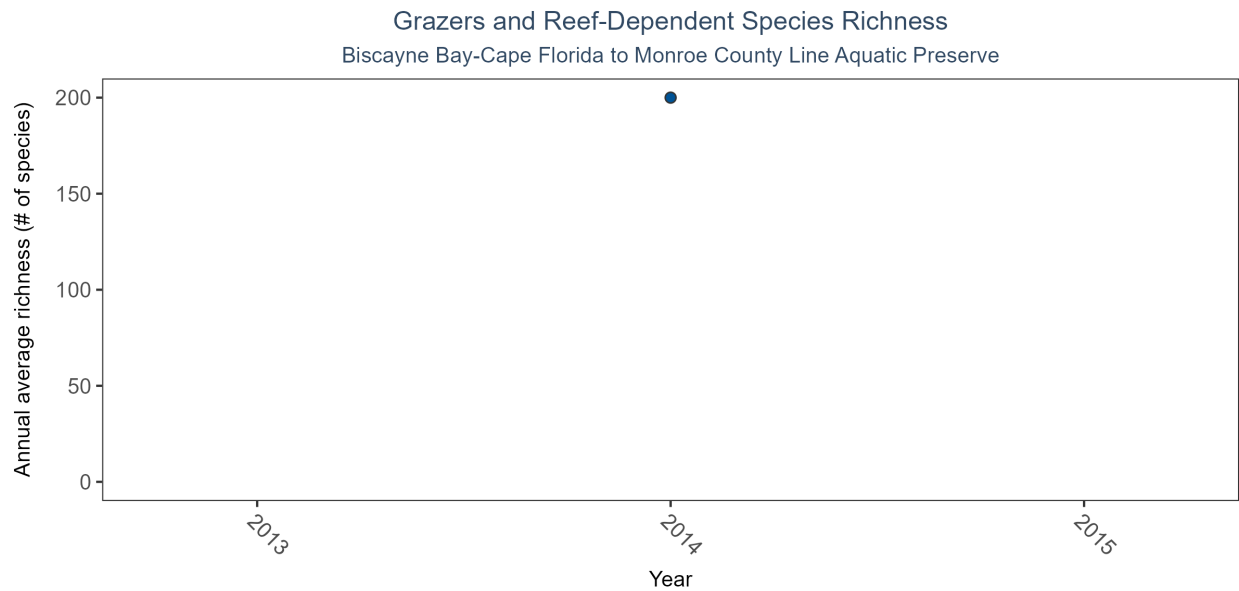
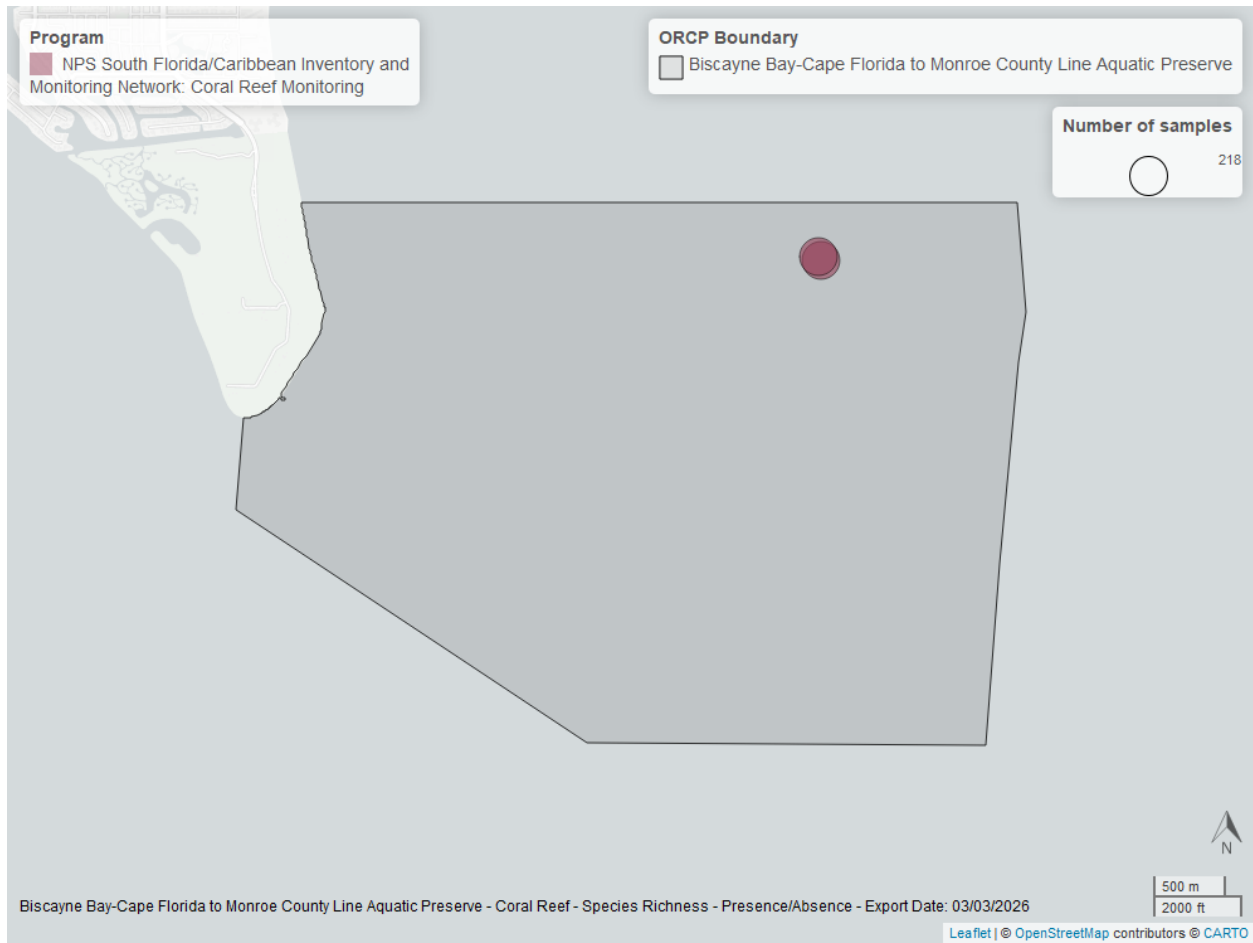


Figure 1: Line graph of annual average species richness of grazers and reef-dependent species over time. If the time series included more than one year of observations, a line connects the data points for visualization.

Table 1: Coral Species Richness - Biscayne Bay-Cape Florida to Monroe County Line Aquatic Preserve

N-Years	EarliestYear	LatestYear	N-Data	Min	Max	Median	Mean	StDev	Year-MinRichness	Year-MaxRichness
1	2014	2014	2	200	200	200	200	0	2014	2014

In the year 2014, 200 taxa were observed based on 2 observations.



Coupon Bight Aquatic Preserve

Table 2: Coral Species Richness - Coupon Bight Aquatic Preserve

N-Years	EarliestYear	LatestYear	N-Data	Min	Max	Median	Mean	StDev	Year-MinRichness	Year-MaxRichness
14	1999	2016	49	200	284	281	266.67	30.69	2014	2012

The median annual number of taxa was 281 based on 49 observations collected between 1999 and 2016.

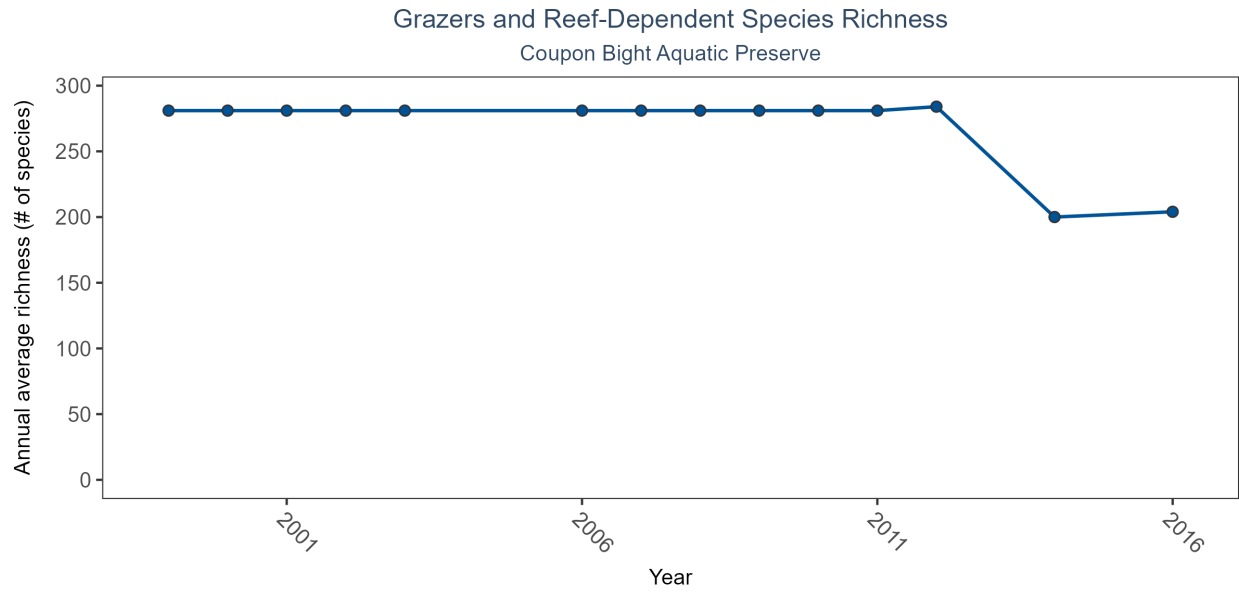
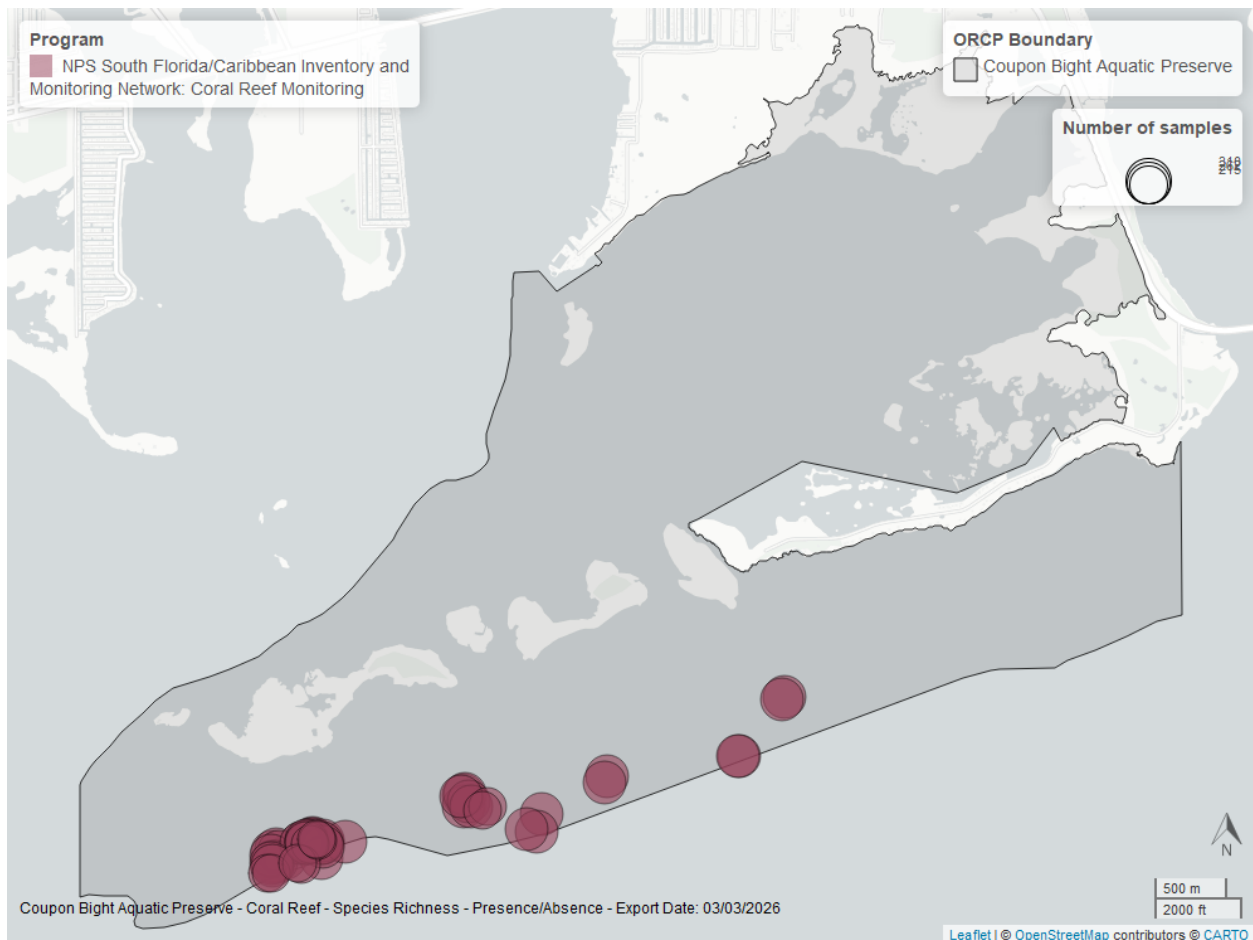


Figure 2: Line graph of annual average species richness of grazers and reef-dependent species over time. If the time series included more than one year of observations, a line connects the data points for visualization.



Florida Keys National Marine Sanctuary

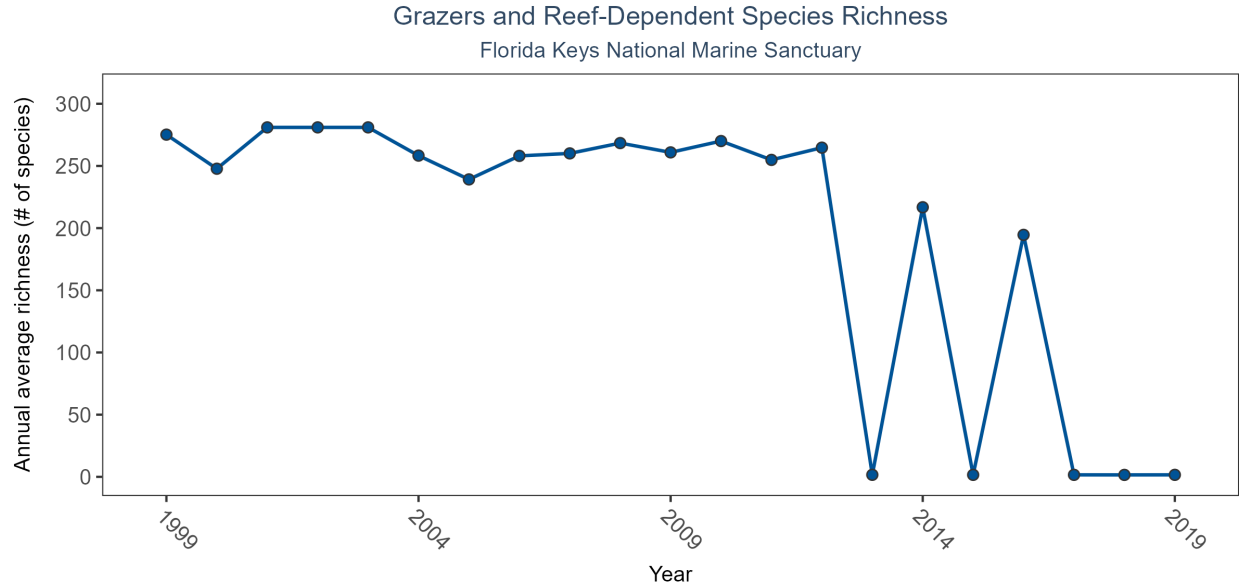
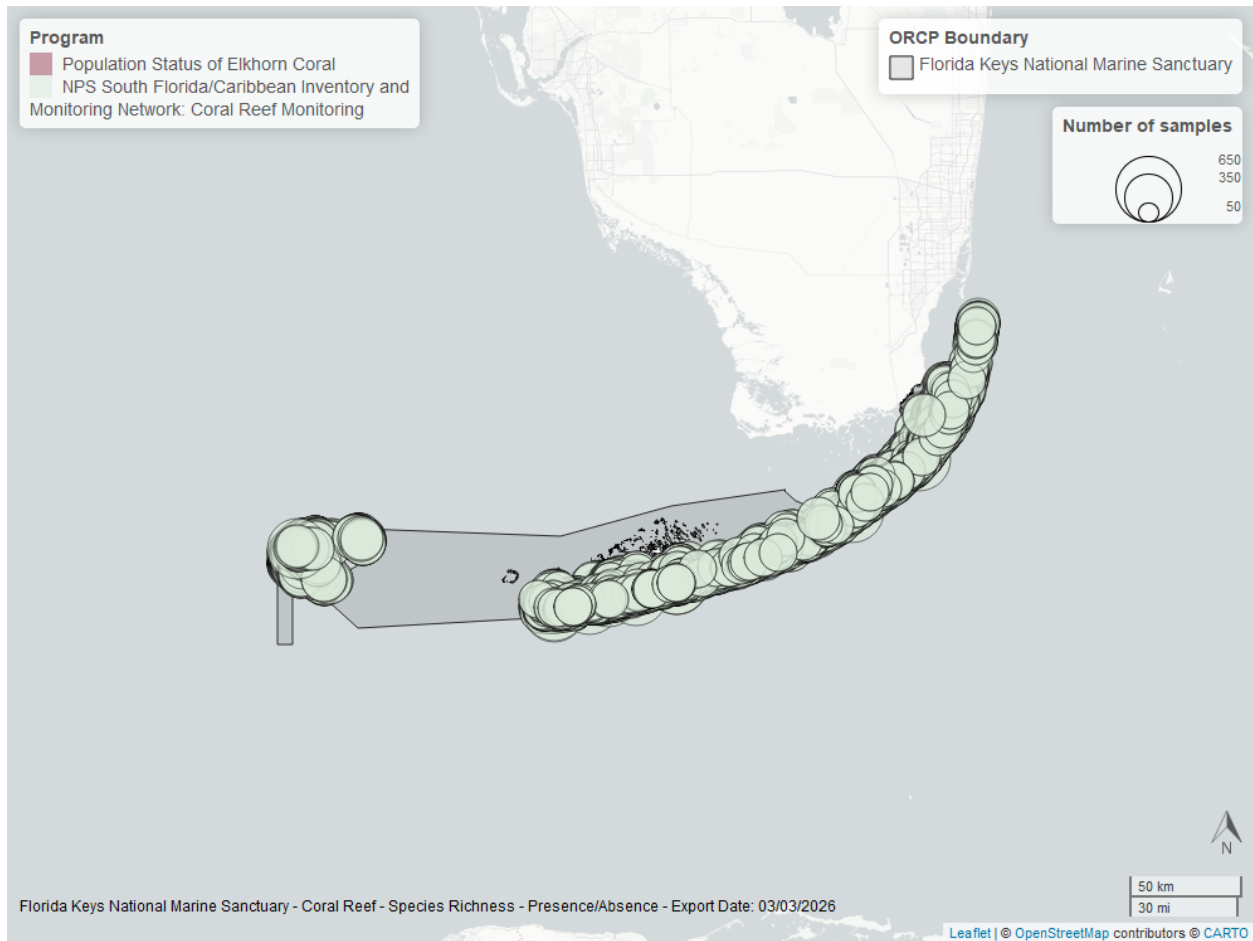


Figure 3: Line graph of annual average species richness of grazers and reef-dependent species over time. If the time series included more than one year of observations, a line connects the data points for visualization.

Table 3: Coral Species Richness - Florida Keys National Marine Sanctuary

N-Years	EarliestYear	LatestYear	N-Data	Min	Max	Median	Mean	StDev	Year-MinRichness	Year-MaxRichness
21	1999	2019	9898	1	302	281	248.15	76.87	2018	2001

The median annual number of taxa was 281 based on 9,898 observations collected between 1999 and 2019.



Kristin Jacobs Coral Aquatic Preserve

Table 4: Coral Species Richness - Kristin Jacobs Coral Aquatic Preserve

N-Years	EarliestYear	LatestYear	N-Data	Min	Max	Median	Mean	StDev	Year-MinRichness	Year-MaxRichness
23	1992	2017	3269	1	302	294	218.45	108.4	1992	2014

The median annual number of taxa was 294 based on 3,269 observations collected between 1992 and 2017.

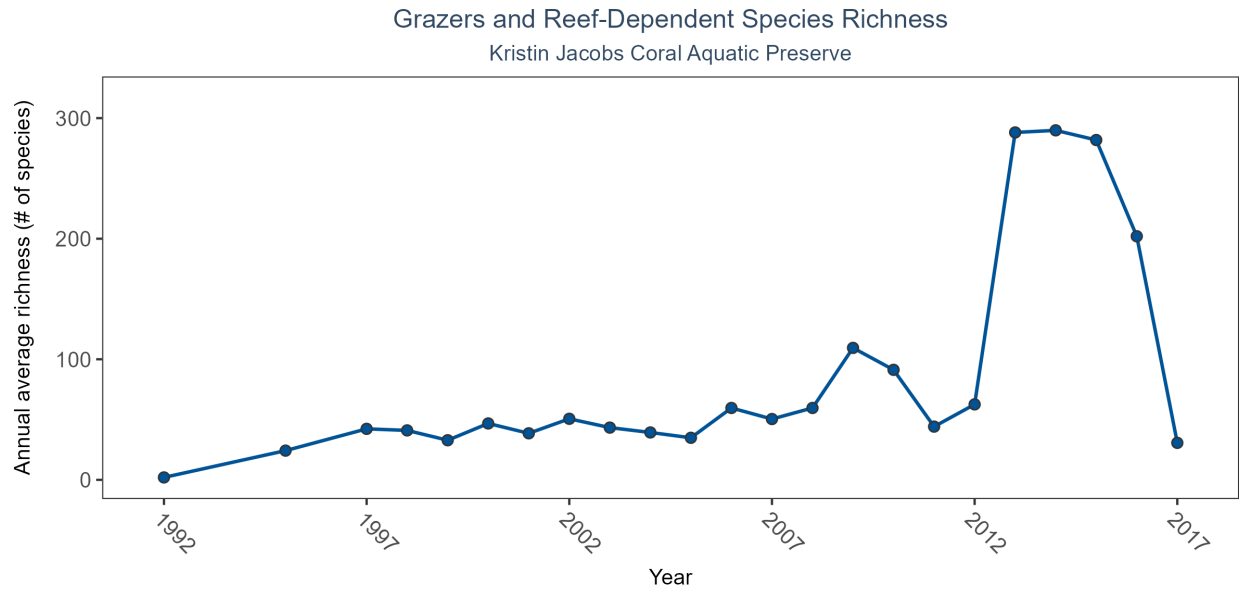
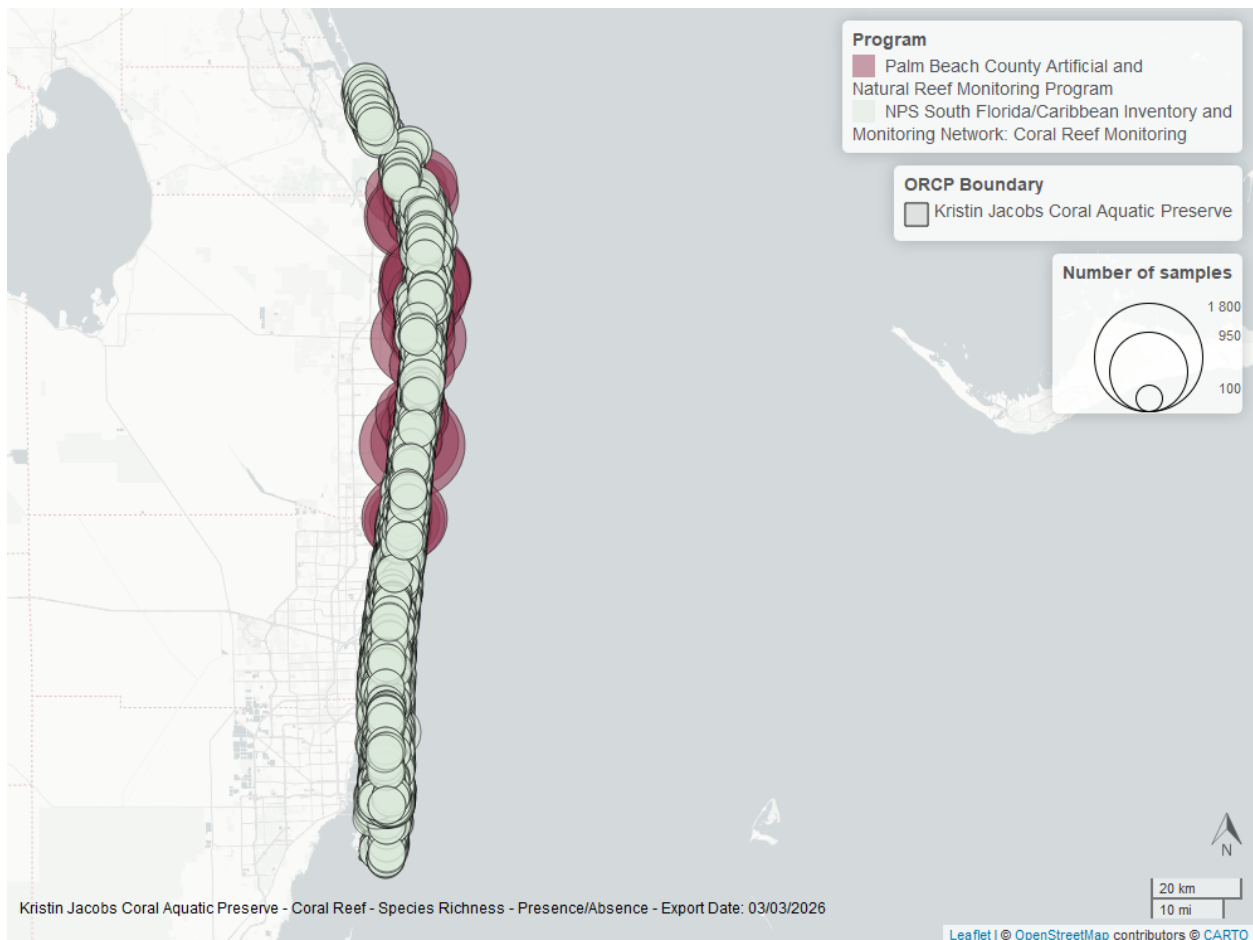


Figure 4: Line graph of annual average species richness of grazers and reef-dependent species over time. If the time series included more than one year of observations, a line connects the data points for visualization.



Libraries and Settings

Loads libraries used in the script. The inclusion of `scipen` option limits how frequently R defaults to scientific notation. Sets default settings for displaying warning and messages in created document, and sets figure dpi.

```
library(knitr)
library(data.table)
library(dplyr)
library(lubridate)
library(ggplot2)
library(ggpubr)
library(scales)
library(EnvStats)
library(tidyr)
library(kableExtra)
library(glue)
library(grid)
library(stringr)
options(scipen=999)
knitr::opts_chunk$set(
  warning=FALSE,
  message=FALSE,
  echo=FALSE
)
options(knitr.kable.NA = '-')
```

File Import

Imports file that is determined in the `Coral_SpeciesRichness_ReportRender.R` script.

The command `fread` is used because of its improved speed while handling large data files. Only columns that are used by the script are imported from the file, and are designated in the `select` input.

The script then gets the name of the parameter as it appears in the data file and units of the parameter.

The latest version of Coral data is available at: <https://usf.box.com/s/8hyj2ur5arothlifglisnq2gxisjzbdg>

The file being used for the analysis is: **All_CORAL_Parameters-2026-Mar-06.txt**

```
# Gets the files with the file names containing the desired parameter
file_in <- list.files(seacar_data_location, pattern="All_CORAL", full=TRUE)

# Reads in data file using fread
data <- fread(file_in, sep="|", header=TRUE, stringsAsFactors=FALSE,
              na.strings=c("NULL", "", "NA"))

cat(paste("The data file(s) used:", file_short, sep="\n"))
```

Data Filtering

The processing and filtering that is done to the data is as follows:

1. Only take data rows that are Presence/Absence measurements for SpeciesGroup1 values “Grazers and reef dependent species” and “Reef fish”
2. Set parameter names to **Species Richness**
3. Sets units

4. Removes rows that contains NA values in ManagedAreaName, GenusName, SpeciesName, Month, Year, SpeciesGroup1, and removes invasive species data
5. Sets ResultValue to be numeric values and removes rows where presence is 0
6. Removes duplicates (MADup==1)
7. Combines genus and species names
8. Corrects some managed area names to match what is being used with other habitats
9. Creates species richness data set
 - Grouped based on common ManagedAreaName, ProgramID, ProgramName, ProgramLocationID, and SampleDate
 - SpeciesRichness determined based on the number of unique species (gensp) in each group
10. Merges data with managed area data to determine correct AreaID
11. Writes to file with “_UsedData” file name to indicate what data was used for species richness.

```
# Only keep data for Presence of grazers and reef-dependent species
data <- data[ParameterName=="Presence/Absence" &
             SpeciesGroup1 %in% c("Grazers and reef dependent species",
                                   "Reef fish"), ]

# Create ParameterName Column
data$ParameterName <- "Species Richness"
parameter <- "Species Richness"
title_param <- "Species Richness - Grazers and Reef-Dependent Species"

# Sets units for species richness
unit <- "# of species"
data$ParameterUnits <- unit

# Remove rows with missing ManagedAreaName
data <- data[!is.na(data$ManagedAreaName),]
data <- data[data$ManagedAreaName!="NA",]
# Remove rows with missing GenusName
data <- data[!is.na(data$GenusName),]
# Remove rows with missing SpeciesName
data <- data[!is.na(data$SpeciesName),]
# Remove rows with missing Months
data <- data[!is.na(data$Month),]
# Remove rows with missing Years
data <- data[!is.na(data$Year),]
# Remove rows with missing SpeciesGroup1
data <- data[!is.na(data$SpeciesGroup1),]
# Set ResultValue to be a number value
data$ResultValue <- as.numeric(data$ResultValue)
# Remove rows where ResultValue is 0 and missing
data <- data[data$ResultValue!=0,]
data <- data[!is.na(data$ResultValue),]
# Remove duplicate rows
data <- data[data$MADup==1,]
# Create variable that combines the genus and species name
data$gensp <- paste(data$GenusName, data$SpeciesName, sep=" ")

# Create Species Richness values for groups of unique combinations of
# ManagedAreaName, ProgramID, ProgramName, ProgramLocationID, and SampleDate.
data <- data[data$ResultValue==1] %>%
  group_by(AreaID, ManagedAreaName, ProgramID, ProgramName, ProgramLocationID,
```

```

        SampleDate) %>%
  summarise(ParameterName=parameter,
            Year=unique(Year), Month=unique(Month),
            SpeciesRichness=length(unique(gensp)))
setDT(data)

# Writes this data that is used by the rest of the script to a text file
fwrite(data, paste0(out_dir, "/Coral_", param_file, "_UsedData.txt"),
       sep="|")

# Makes sure SampleDate is being stored as a Date object
data$SampleDate <- as.Date(data$SampleDate)

# Creates a variable with the names of all the managed areas that contain
# species observations
coral_sr_MA_Include <- unique(data$ManagedAreaName[!is.na(data$SpeciesRichness)])

# Puts the managed areas in alphabetical order
coral_sr_MA_Include <- coral_sr_MA_Include[order(coral_sr_MA_Include)]

# Determines the number of managed areas used
n <- length(coral_sr_MA_Include)

```

Managed Area Statistics

Gets summary statistics for each managed area. Uses piping from dplyr package to feed into subsequent steps. The following steps are performed:

1. Group data that have the same `ManagedAreaName`, `Year`, and `Month`.
 - Second summary statistics do not use the `Month` grouping and are only for `ManagedAreaName` and `Year`.
 - Third summary statistics do not use `Year` grouping and are only for `ManagedAreaName` and `Month`
 - Fourth summary statistics are only grouped based on `ManagedAreaName`
 - Determines the years that the minimum and maximum species richness occurred
2. For each group, provide the following information: Parameter Name (`ParameterName`), Number of Entries (`N_Data`), Lowest Value (`Min`), Largest Value (`Max`), Median, Mean, Standard Deviation, and a list of all Programs included in these measurements.
3. Sort the data in ascending (A to Z and 0 to 9) order based on `ManagedAreaName` then `Year` then `Month`
4. Write summary stats to a pipe-delimited .txt file in the output directory
 - [Coral Species Richness Output Files in SEACAR GitHub](https://github.com/FloridaSEACAR/SEACAR_Trend_Analyses/tree/main/Coral/output/SpeciesRichness) (https://github.com/FloridaSEACAR/SEACAR_Trend_Analyses/tree/main/Coral/output/SpeciesRichness)

```

# Create summary statistics for each managed area based on Year and Month
# intervals.
MA_YM_Stats <- data %>%
  group_by(AreaID, ManagedAreaName, Year, Month) %>%
  summarize(ParameterName=parameter,
            N_Data=length(na.omit(SpeciesRichness)),
            Min=min(SpeciesRichness),
            Max=max(SpeciesRichness),
            Median=median(SpeciesRichness),
            Mean=mean(SpeciesRichness),
            StandardDeviation=sd(SpeciesRichness),
            Programs=paste(sort(unique(ProgramName), decreasing=FALSE),

```

```

        collapse=', '),
    ProgramIDs=paste(sort(unique(ProgramID), decreasing=FALSE),
        collapse=', '),
    .groups = "keep")
# Puts the data in order based on ManagedAreaName, Year, then Month
MA_YM_Stats <- as.data.table(MA_YM_Stats[order(MA_YM_Stats$ManagedAreaName,
        MA_YM_Stats$Year,
        MA_YM_Stats$Month), ])

# Writes summary statistics to file
fwrite(MA_YM_Stats, paste0(out_dir, "/Coral_", param_file,
        "_MA_MMY_Stats.txt"), sep="|")
# Removes variable storing data to improve computer memory
rm(MA_YM_Stats)

# Create summary statistics for each managed area based on Year intervals
MA_Y_Stats <- data %>%
    group_by(AreaID, ManagedAreaName, Year) %>%
    summarize(ParameterName=parameter,
        N_Data=length(na.omit(SpeciesRichness)),
        Min=min(SpeciesRichness),
        Max=max(SpeciesRichness),
        Median=median(SpeciesRichness),
        Mean=mean(SpeciesRichness),
        StandardDeviation=sd(SpeciesRichness),
        Programs=paste(sort(unique(ProgramName), decreasing=FALSE),
            collapse=', '),
        ProgramIDs=paste(sort(unique(ProgramID), decreasing=FALSE),
            collapse=', '),
        .groups = "keep")
# Puts the data in order based on ManagedAreaName then Year
MA_Y_Stats <- as.data.table(MA_Y_Stats[order(MA_Y_Stats$ManagedAreaName,
        MA_Y_Stats$Year), ])

# Writes summary statistics to file
fwrite(MA_Y_Stats, paste0(out_dir, "/Coral_", param_file,
        "_MA_Yr_Stats.txt"), sep="|")

# Create summary statistics for each managed area based on Month intervals.
MA_M_Stats <- data %>%
    group_by(AreaID, ManagedAreaName, Month) %>%
    summarize(ParameterName=parameter,
        N_Data=length(na.omit(SpeciesRichness)),
        Min=min(SpeciesRichness),
        Max=max(SpeciesRichness),
        Median=median(SpeciesRichness),
        Mean=mean(SpeciesRichness),
        StandardDeviation=sd(SpeciesRichness),
        Programs=paste(sort(unique(ProgramName), decreasing=FALSE),
            collapse=', '),
        ProgramIDs=paste(sort(unique(ProgramID), decreasing=FALSE),
            collapse=', '),
        .groups = "keep")
# Puts the data in order based on ManagedAreaName then Month
MA_M_Stats <- as.data.table(MA_M_Stats[order(MA_M_Stats$ManagedAreaName,

```

```

MA_M_Stats$Month), ]))

# Writes summary statistics to file
fwrite(MA_M_Stats, paste0(out_dir, "/Coral_", param_file,
                           "_MA_Mo_Stats.txt"), sep="|")
# Removes variable storing data to improve computer memory
rm(MA_M_Stats)

# Create summary overall statistics for each managed area.
MA_Ov_Stats <- data %>%
  group_by(AreaID, ManagedAreaName) %>%
  summarize(ParameterName=parameter,
             N_Years=length(unique(na.omit(Year))),
             EarliestYear=min(Year),
             LatestYear=max(Year),
             N_Data=length(na.omit(SpeciesRichness)),
             Min=min(SpeciesRichness),
             Max=max(SpeciesRichness),
             Median=median(SpeciesRichness),
             Mean=mean(SpeciesRichness),
             StandardDeviation=sd(SpeciesRichness),
             Programs=paste(sort(unique(ProgramName), decreasing=FALSE),
                             collapse=', '),
             ProgramIDs=paste(sort(unique(ProgramID), decreasing=FALSE),
                               collapse=', '),
             .groups = "keep")
# Puts the data in order based on ManagedAreaName
MA_Ov_Stats <- as.data.table(MA_Ov_Stats[order(MA_Ov_Stats$ManagedAreaName), ])
# Creates Year_MinRichness and Year_MaxRichness columns
MA_Ov_Stats$Year_MinRichness <- NA
MA_Ov_Stats$Year_MaxRichness <- NA

# Loops through each ManagedAreaName.
# Determines what year the minimum and maximum species richness occurred
for(m in 1:nrow(MA_Ov_Stats)){
  # Stores ManagedAreaName for this row
  ma <- MA_Ov_Stats$ManagedAreaName[m]

  # Skips to next row if there are no data for this combination
  if(MA_Ov_Stats$N_Data[m]==0){
    next
  }
  # Gets subset of data from MA_Y_Stats (yearly summary stats) with this
  # ManagedAreaName
  ds <- MA_Y_Stats[ManagedAreaName==ma,]
  # Gets the minimum and maximum Mean (yearly averages)
  min <- min(ds$Mean)
  max <- max(ds$Mean)
  #Determines what years those minimum and maximum values occurred
  year_min <- ds[Mean==min, Year]
  year_max <- ds[Mean==max, Year]
  # Stores the occurrence years of the minimum and maximum into the overall
  # stats for this row
  MA_Ov_Stats$Year_MinRichness[m] <- year_min

```

```

MA_Ov_Stats$Year_MaxRichness[m] <- year_max
}
# Replaces blank ProgramIDs with NA (missing values)
MA_Ov_Stats$ProgramIDs <- replace(MA_Ov_Stats$ProgramIDs,
                                  MA_Ov_Stats$ProgramIDs=="", NA)
MA_Ov_Stats$Programs <- replace(MA_Ov_Stats$Programs,
                                 MA_Ov_Stats$Programs=="", NA)
# Write overall statistics to file
fwrite(MA_Ov_Stats, paste0(out_dir, "/Coral_", param_file,
                             "_MA_Overall_Stats.txt"), sep="|")
# Removes entries from the overall statistics that do not have data.
# Based on presence or absence of EarliestYear
MA_Ov_Stats <- MA_Ov_Stats[!is.na(MA_Ov_Stats$EarliestYear), ]

```

Appendix I: Managed Area Species Richness

The plots shown here are the species richness for each managed area with a yearly average.

1. Set common plot theme.
2. Determine the earliest and latest year of the data to create x-axis scale and intervals
3. Determine the upper and lower limit of the plot for better y-axis labels
4. Add the plot line
5. Set the plot type as a point plot with the size of the points
6. Create the title, x-axis, y-axis
7. Set the y and x limits
8. Apply common plot theme
9. Add table with summary statistics below each figure
 - Numerical non-integer values are rounded to 2 decimal places
 - StandardDeviation is renamed StDev for space reasons
10. Create file name to save figure
11. Save figure as png file
 - Coral Species Richness Figures in SEACAR GitHub (https://github.com/FloridaSEACAR/SEACAR_Trend_Analyses/tree/main/Coral/output/SpeciesRichness/Figures)

```

# Defines standard plot theme: black and white, no major or minor grid lines,
# Arial font. Title is centered, size 12, and blue (hex coded). Subtitle is
# centered, size 10, and blue (hex coded). Legend title is size 10 and the
# legend is left-justified. X-axis title is size 10 and the margins are padded
# at the top and bottom to give more space for angled axis labels. Y-axis title
# is size 10 and margins are padded on the right side to give more space for
# axis labels. Axis labels are size 10 and the x-axis labels are rotated -45
# degrees with a horizontal justification that aligns them with the tick mark
plot_theme <- theme_bw() +
  theme(panel.grid.major = element_blank(),
        panel.grid.minor = element_blank(),
        text=element_text(family="Arial"),
        plot.title=element_text(hjust=0.5, size=12, color="#314963"),
        plot.subtitle=element_text(hjust=0.5, size=10, color="#314963"),
        legend.title=element_text(size=10),
        legend.text = element_text(hjust=0),
        axis.title.x = element_text(size=10, margin = margin(t = 5, r = 0,
                                                              b = 10, l = 0)),
        axis.title.y = element_text(size=10, margin = margin(t = 0, r = 10,

```

```

axis.text=element_text(size=10),
axis.text.x=element_text(angle = -45, hjust = 0))

# Color palette for SEACAR
color_palette <- c("#005396", "#0088B1", "#00ADAE", "#65CCB3", "#AEE4C1",
                  "#FDEBA8", "#F8CD6D", "#F5A800", "#F17B00")

# Loop that cycles through each managed area with data
if(n==0){
  # Prints a statement if there are no managed areas with appropriate data
  print("There are no monitoring locations that qualify.")
} else {
  for (i in 1:n) {
    ma_i <- coral_sr_MA_Include[i]
    # Gets data for target managed area
    plot_data <- MA_Y_Stats[ManagedAreaName==ma_i, ]
    # Determines most recent year with available data for managed area
    t_max <- max(MA_Ov_Stats[ManagedAreaName==ma_i, LatestYear])
    # Determines earliest recent year with available data for managed area
    t_min <- min(MA_Ov_Stats[ManagedAreaName==ma_i, EarliestYear])
    # Determines how many years of data are present
    t <- t_max-t_min

    # Creates break intervals for plots based on number of years of data
    if(t>=30){
      # Set breaks to every 10 years if more than 30 years of data
      brk <- -10
    }else if(t<30 & t>=10){
      # Set breaks to every 5 years if between 30 and 10 years of data
      brk <- -5
    }else if(t<10 & t>=4){
      # Set breaks to every 2 years if between 10 and 4 years of data
      brk <- -2
    }else if(t<4 & t>=2){
      # Set breaks to every year if between 4 and 2 years of data
      brk <- -1
    }else if(t<2){
      # Set breaks to every year if less than 2 years of data
      brk <- -1
      # Sets t_max to be 1 year greater and t_min to be 1 year lower
      # Forces graph to have at least 3 tick marks
      t_max <- t_max+1
      t_min <- t_min-1
    }
    # Determine range of data values for the managed area
    y_range <- max(plot_data$Mean) - min(plot_data$Mean)

    # Determines lower bound of y-axis based on data range. Set based on
    # relation of data range to minimum value. Designed to set lower boundary
    # to be 10% of the data range below the minimum value
    y_min <- if(min(plot_data$Mean)-(0.1*y_range)<0){
      # If 10% of the data range below the minimum value is less than 0,

```

```

# set as 0
y_min <- 0
} else {
  # Otherwise set minimum bound as 10% data range below minimum value
  y_min <- min(plot_data$Mean)-(0.1*y_range)
}

# Sets upper bound of y-axis to be 10% of the data range above the
# maximum value.
y_max <- max(plot_data$Mean)+(0.1*y_range)

# Creates plot object using plot_data.
# Data is plotted as symbols with connected lines.
p1 <- ggplot(data=plot_data) +
  geom_line(aes(x=Year, y=Mean), color=color_palette[1],
            size=0.75, alpha=1) +
  geom_point(aes(x=Year, y=Mean), fill=color_palette[1],
            shape=21, size=2, color="#333333", alpha=1) +
  labs(title="Grazers and Reef-Dependent Species Richness",
        subtitle=ma_i,
        x="Year", y="Richness (# of species)") +
  scale_x_continuous(limits=c(t_min-0.25, t_max+0.25),
                    breaks=seq(t_max, t_min, brk)) +
  scale_y_continuous(limits=c(y_min, y_max),
                    breaks=pretty_breaks(n=5)) +
  plot_theme
# Sets file name of plot created
outname <- paste0("Coral_", param_file, "_", gsub(" ", "", ma_i),
                 ".png")
# Saves plot as a png image
png(paste0(out_dir, "/Figures/", outname),
    width = 8,
    height = 4,
    units = "in",
    res = 200)
print(p1)
dev.off()

# Creates a data table object to be shown underneath plots in report
ResultTable <- MA_Ov_Stats[ManagedAreaName==ma_i,]
# Removes location, and parameter information because it is in plot
# labels
ResultTable <- ResultTable[, -c("AreaID", "ManagedAreaName",
                               "ProgramIDs", "Programs", "ParameterName")]
# Renames StandardDeviation to StDev to save horizontal space
ResultTable <- ResultTable %>%
  rename("StDev"="StandardDeviation")
# Converts all non-integer values to 2 decimal places for space
ResultTable$Min <- round(ResultTable$Min, digits=2)
ResultTable$Max <- round(ResultTable$Max, digits=2)
ResultTable$Median <- round(ResultTable$Median, digits=2)
ResultTable$Mean <- round(ResultTable$Mean, digits=2)
ResultTable$StDev <- round(ResultTable$StDev, digits=2)

```

```

# Stores as plot table object
t1 <- ggtexttable(ResultTable, rows = NULL,
                  theme=ttheme(base_size=7))
# Combines plot and table into one figure
print(ggarrange(p1, t1, ncol=1, heights=c(0.85, 0.15)))

# Add extra space at the end to prevent the next figure from being too
# close. Does not add space after last plot
if(i!=n){
  cat("\n \n \n \n")
}
}
}

```